

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12387958
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Wiseman; Brian et al.

Perimeter frame for spacing reticle pods

Abstract

A spacer for parts of a reticle pod includes spacer contacts configured to contact alignment features of a cover and a baseplate of the reticle pod such that the cover and baseplate are separated by a gap. The spacer includes a frame connecting at least two of the spacer contacts. The frame can include a spring arm allowing deformation such that the spacer contacts can be moved relative to one another to facilitate interfacing with the alignment features. The spacer contacts can be configured to surround pins included in the alignment features. The spacer contacts can be configured to be inserted into lead-in portions of channels included in the alignment features.

Inventors: Wiseman; Brian (Glencoe, MN), Raschke; Russ V. (Chanhassen, MN), Magoon; Paul J. (Colorado Springs, CO), Wang; Huaping (Eden Prairie, MN)

Applicant: ENTEGRIS, INC. (Billerica, MA)

Family ID: 90244402

Assignee: ENTEGRIS, INC. (Billerica, MA)

Appl. No.: 18/370570

Filed: September 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240096670 A1	Mar. 21, 2024

Related U.S. Application Data

us-provisional-application US 63408451 20220920

Publication Classification

Int. Cl.: H01L21/673 (20060101)

U.S. Cl.:

CPC H01L21/67359 (20130101); H01L21/67386 (20130101);

Field of Classification Search

CPC: G03F (1/38); G03F (1/66); H01L (21/673); H01L (21/67359); H01L (21/67386); H01L (21/67)

USPC: 206/316.1

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6216873	12/2000	Fosnight	N/A	N/A
10281815	12/2018	Hsueh	N/A	H01L 21/67359
D889941	12/2019	McCoy	N/A	N/A
D902179	12/2019	Miller et al.	N/A	N/A
D964948	12/2021	Chuang et al.	N/A	N/A
D964949	12/2021	Chuang et al.	N/A	N/A
D964950	12/2021	Chuang et al.	N/A	N/A
11442370	12/2021	Chuang	N/A	H01L 21/67386
2008/0254377	12/2007	Chen	N/A	N/A
2010/0038264	12/2009	Balix	N/A	N/A
2021/0047094	12/2020	Chuang	N/A	N/A
2023/0333463	12/2022	Wang et al.	N/A	N/A
2024/0096670	12/2023	Wiseman et al.	N/A	N/A
2024/0160098	12/2023	Raschke et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2013224169	12/2012	JP	N/A
I666510	12/2018	TW	N/A
D210557	12/2020	TW	N/A

Primary Examiner: Gehman; Bryon P

Background/Summary

FIELD

(1) This disclosure is directed to reticle pods including spacers and spacers connected by frames for spacing sealing surfaces of reticle pods apart from one another.

BACKGROUND

(2) Reticle containers can include hard sealing surfaces. When reticle containers are shipped, they are typically shipped in their complete assembled state. During shipping, the sealing surfaces may

be subject to shock and/or vibration in excess of ordinary handling and operations for the reticle pod, leading to wear, damage, and particle generation that can affect the cleanliness or other performance of the reticle pod.

SUMMARY

(3) This disclosure is directed to reticle pods including spacers and spacers connected by frames for spacing sealing surfaces of reticle pods apart from one another.

(4) By providing spacers that interface with alignment features of a reticle container, such as an inner pod of a reticle pod, a gap can be maintained between the sealing surfaces while the reticle pod is shipped in the assembled state. The presence of such a gap can reduce or eliminate wear or damage resulting from shocks or vibration experienced during shipping of the reticle container. The gap can be sized such that the reticle container can still be completely assembled, for example having an inner pod be contained within the outer pod of the complete reticle container during such shipping. This allows reticle containers to be shipped with reduced risk of damage or particle generation, even when shipped as complete containers. Further, the contacts of the spacer can be joined by a frame to facilitate handling. The frame can be configured such that it runs outside a perimeter of the sealing surfaces of the reticle container, reducing the likelihood of wear particles from the spacer from causing cleanliness issues in the reticle pod.

(5) In an embodiment, a spacer for a reticle pod includes a plurality of spacing contacts. Each of the spacing contacts is configured to contact a first reticle pod segment at a first reticle pod alignment feature and to contact a second reticle pod segment at a second reticle pod alignment feature. The first reticle pod alignment feature is separate from a first sealing surface of the first reticle pod segment, and the second reticle pod alignment feature is separate from a second sealing surface of the second reticle pod segment. The spacer includes a frame joining at least two of the plurality of spacing contacts. The frame is configured to be outside the first sealing surface and the second sealing surface in plan view of the reticle pod. The spacing contacts are configured to contact the first reticle pod segment and the second reticle pod segment such that the first reticle pod segment and the second reticle pod segment are separated by a gap.

(6) In an embodiment, the plurality of spacing contacts includes at least three of the spacing contacts.

(7) In an embodiment, the frame joins all of the spacing contacts of the plurality of spacing contacts.

(8) In an embodiment, the frame comprises at least one spring arm configured to allow deflection of the frame such that a distance between the plurality of spacing contacts is varied.

(9) In an embodiment, each of the spacing contacts includes an inner surface defining an opening configured to receive the first reticle pod alignment feature, and an outer surface configured to be received within the second reticle pod alignment feature.

(10) In an embodiment, the gap has a width in a range from 20 μm to 1 mm.

(11) In an embodiment, the spacer includes at least one of polycarbonate and a polyolefin.

(12) In an embodiment, a reticle pod includes an inner pod including a cover and a baseplate. The cover includes a cover sealing surface and a plurality of cover alignment features separate from the cover sealing surface. The baseplate includes a baseplate sealing surface and a plurality of baseplate alignment features separate from the baseplate sealing surface. The reticle pod includes a spacer. The spacer includes a plurality of spacing contacts. Each of the spacing contacts is configured to contact one of the plurality of cover alignment features and to contact one of the plurality of baseplate alignment features. The spacer includes a frame joining at least two of the plurality of spacing contacts, the frame configured to be outside the cover sealing surface and the baseplate sealing surface in plan view of the reticle pod. The spacing contacts are configured to contact the cover and the baseplate such that the cover and the baseplate are separated by a gap.

(13) In an embodiment, the plurality of spacing contacts includes at least three of the spacing contacts.

- (14) In an embodiment, the frame joins all of the spacing contacts of the plurality of spacing contacts.
- (15) In an embodiment, the frame comprises at least one spring arm configured to allow deflection of the frame.
- (16) In an embodiment, the cover alignment feature is a pin and the baseplate alignment feature is a channel configured to receive the pin, the channel including a lead-in.
- (17) In an embodiment, each of the spacing contacts includes an inner surface defining an opening configured to receive the pin, and an outer surface configured to be received within the lead-in.
- (18) In an embodiment, the gap has a width in a range from 20 μm to 1 mm.
- (19) In an embodiment, the reticle pod further includes an outer pod including a pod dome and a pod door, the outer pod configured to accommodate the inner pod.
- (20) In an embodiment, a method of preparing a reticle pod for shipment includes assembling an inner pod including a cover, a baseplate, and a spacer provided between the cover and the baseplate. The spacer includes a plurality of spacing contacts contacting the cover at cover alignment features. The spacer contacts the baseplate at baseplate alignment features. The spacer includes a frame joining at least two of the plurality of spacing contacts. The cover and the baseplate are separated by a gap.
- (21) In an embodiment, the frame includes a deflection arm, and assembling the inner pod includes deflecting the frame at the deflection arm to align at least one of the plurality of spacing contacts with one of the cover alignment features or one of the baseplate alignment features.
- (22) In an embodiment, the gap has a width in a range from 20 μm to 1 mm.
- (23) In an embodiment, the cover alignment features are each a pin, the baseplate alignment features are each a channel including a lead-in, and assembling the inner pod includes passing each of the pins through one of the spacing contacts, and inserting each of the spacing contacts into one of the lead-ins.
- (24) In an embodiment, the method further includes placing the inner pod into an outer pod, the outer pod including a pod dome and a pod door, joining the pod dome to the pod door to enclose the inner pod, and shipping the reticle pod from a packing location to a destination location while the pod dome is joined to the pod door.
-

Description

DRAWINGS

- (1) FIG. 1 shows a reticle container according to an embodiment.
- (2) FIG. 2A shows a sectional view of an inner pod of a reticle container according to an embodiment.
- (3) FIG. 2B shows a plan view of the cover and the spacer of FIG. 2A.
- (4) FIG. 3 shows a spacer according to an embodiment.
- (5) FIG. 4 shows a spacer according to an embodiment.
- (6) FIG. 5 shows a reticle pod according to an embodiment.

DETAILED DESCRIPTION

- (7) This disclosure is directed to reticle pods including spacers and spacers connected by frames for spacing sealing surfaces of reticle pods apart from one another.
- (8) FIG. 1 shows a reticle container according to an embodiment. Reticle container **100** includes outer pod **102** including pod dome **104** and pod door **106**. Reticle container **100** also includes inner pod **108**, including cover **110** and baseplate **112**. Reticle container **100** further includes spacer **114**.
- (9) Outer pod **102** forms an exterior of reticle container **100**. Outer pod **102** is configured to accommodate inner pod **108**. When reticle container **100** is shipped to a destination or stored at a location, the reticle container can be assembled with the inner pod **108** contained within the outer

pod **102**. The outer pod **102** can be composed of pod dome **104** and pod door **106**.

(10) Inner pod **108** is an inner pod of reticle container **100**, configured to accommodate a reticle. The inner pod includes cover **110** and baseplate **112**. Cover **110** and baseplate **112** can be configured to contact one another during typical use, for example at corresponding sealing surfaces provided on each of cover **110** and baseplate **112**. Cover **110** and baseplate **112** can be made of metal materials, and further can have metal plating, for example at least at some of the sealing surfaces. The cover **110** and baseplate **112** can be spaced apart from one another at other times, for example during storage or shipment, for example using one or more of spacer **114**.

(11) Spacer **114** can be included in reticle container **100** to space cover **110** and baseplate **112** apart from one another when inner pod **108** is assembled. The spacer **114** can include one or more spacing contacts configured to engage with features included in cover **110** and baseplate **112**. The features can be any suitable features disposed outwards of sealing surfaces included in the cover **110** and baseplate **112**. The features can be between an outer perimeter of the sealing surfaces and an outer perimeter of the respective cover **110** or baseplate **112**. Non-limiting examples of such features include alignment features, ledges, shelves, projections, flanges, combinations thereof, and the like provided on the cover **110** and the baseplate **112**. The features that the spacer **114** engage with are disposed on parts of the cover **110** and/or baseplate **112** that are outside of sealing surfaces. Outside of the sealing surfaces means that the features engaged with and the position of the frame of spacer **114** do not overlap with the sealing surfaces of cover **110** or baseplate **112** in plan view when the cover **110** and baseplate **112** are assembled with spacer **114**. The spacer **114** can contact the cover **110** and the baseplate **112** such that the respective sealing surfaces remain separated by a gap. In an embodiment, the gap can be maintained when the cover **110** and the baseplate **112** are assembled with spacer **114** in place and placed into the outer pod **102**. When inner pod **108** is placed in outer pod **102** and outer pod **102** is closed, the gap can have a size of between 20 micrometer (μm) and 1 millimeter (mm). In an embodiment, the spacer **114** can include one or more spring members configured to increase a size of the gap when inner pod **108** is not contained in outer pod **102**, relative to the size of the gap maintained when inner pod **108** is contained within the outer pod **102**.

(12) Optionally, cover **110** can include cover flanges **116**. Cover flanges **116** project outwards from sides of the cover **110**. The cover flanges **116** can provide a point where spring members included in spacer **114** can contact the cover **110** to provide the increase in gap size when the cover **110** and baseplate **112** are not contained within outer pod **102**.

(13) FIG. 2A shows a sectional view an inner pod of a reticle container according to an embodiment. Inner pod **200** includes cover **202** including cover alignment features **204** and cover sealing surface **206**. Inner pod **200** also includes baseplate **208** including baseplate alignment features **210** and baseplate sealing surface **212**. Spacer **214** includes spacing contacts **216** and frame **218**.

(14) Cover **202** is a cover of the inner pod **200**. Cover **202** is configured to combine with baseplate **208** to define an internal space capable of accommodating a reticle. The cover **202** can be configured to form an upper portion of the inner pod **200**. The cover **202** includes cover alignment features **204**. The cover alignment features can be any suitable features provided in cover **204** to provide alignment of cover **202** with baseplate **208**, for example by engagement between the cover alignment features **204** with suitable baseplate alignment features **210**. In the embodiment shown in FIG. 2, the cover alignment features **204** include a plurality of pins protruding from the cover **202**. In embodiments, the cover alignment features can instead be recesses or openings configured to receive pins provided as the baseplate alignment features **210** or any other suitable structure for guiding and/or maintaining alignment of cover **202** with baseplate **208**. Cover **202** also includes cover sealing surface **206**. Cover sealing surface **206** is a surface of cover **202** configured to contact the baseplate **208** when inner pod **200** is assembled for typical use such as containing a reticle. The cover sealing surface **206** can be one or more surfaces positioned to correspond to baseplate sealing

surface **212** when inner pod **200** is assembled.

(15) Baseplate **208** forms another portion of inner pod **200**. The baseplate **208** is configured to, when combined with cover **202**, define an internal space capable of accommodating a reticle. Baseplate **208** includes baseplate alignment features **210**. In the embodiment shown in FIG. 2, the baseplate alignment features **210** are holes formed in baseplate **208**, configured to receive the pins that are cover alignment features **204**. The holes include lead-ins having a diameter larger than the diameter of the pins. The baseplate **208** further includes baseplate sealing surface **212**. Baseplate sealing surface **212** is configured to contact cover sealing surface **206** when the inner pod **200** is assembled, for example to contain a reticle. The baseplate sealing surface **212** can oppose the corresponding cover sealing surface **206**.

(16) Spacer **214** includes spacing contacts **216** joined by frame **218**. In the embodiment shown in FIG. 2, spacing contacts **216** are each configured to contact the cover **202** and the baseplate **208** at the respective cover alignment features **204** and baseplate alignment features **210**. It is understood that the position and shape of spacing contacts **216** can be any suitable configuration for engaging the features of the cover **202** and the baseplate **208** outside of the respective cover sealing surface **206** and the baseplate sealing surface **212** so as to space the sealing surfaces **206**, **212** apart from one another. For example, tabs engaging ledges or shelves can be used as the spacing contacts **216** in embodiments contacting the cover **202** and/or the baseplate **208** at such ledges or shelves.

Spacing contacts **216** are configured to contact the cover alignment features **204** and the baseplate alignment features **210** such that cover sealing surfaces **206** and baseplate sealing surfaces **212** are spaced apart from one another by a gap. In the embodiment shown in FIG. 2, the spacing contacts **216** are each ring-shaped and include an inner surface defining a bore configured to receive one of the pins of the cover alignment feature **204**. The spacing contacts **216** each further include an outer surface configured to be received in one of the lead-ins of the holes of the baseplate alignment features **210**. The contact between the spacing contacts **216** and the cover and baseplate alignment features **204**, **210** can hold cover **202** apart from baseplate **208** such that a gap separates the cover sealing surface **206** and the baseplate sealing surface **212**. In an embodiment, the number of spacing contacts **216** corresponds to the number of matched cover and baseplate alignment features **204**, **210** included in inner pod **200**. In an embodiment, the number of spacing contacts **216** is less than the number of matched cover and baseplate alignment features **204**, **210** included in inner pod **200**. The spacer **214** can be installed, for example, for shipping of inner pod **200**, to prevent contact between the cover and baseplate sealing surfaces **206**, **212** during such shipment or storage of inner pod **200**. The spacer **214** can be configured to maintain separation between cover **202** and baseplate **208** even when the inner pod **200** is placed into an outer pod, such as outer pod **102** described above and shown in FIG. 1, and the outer pod is closed. Spacer **214** can include a frame joining two or more of the spacing contacts. Frame **218** can be configured to run outside of the cover sealing surface **206** and/or the baseplate sealing surface **212** with respect to a plan view of the inner pod **200**. By avoiding the areas corresponding to the cover and baseplate sealing surfaces **206**, **212**, the amount of particles generated by spacer **214** and its contact with cover **202** and/or baseplate **208** contacting cover and baseplate sealing surfaces **206**, **212** and the interior space defined by cover **202** and baseplate **208** can be reduced.

(17) While cover alignment features **204** are shown as being pins and baseplate alignment features **210** are shown as channels receiving the pins in FIG. 2, it is understood that pins could be provided as the baseplate alignment features **210** and channels receiving the pins could be provided as the cover alignment features **204**. It is understood that the spacer **214** can be shaped and sized appropriately to maintain the gap between cover and baseplate sealing surfaces **206**, **212** by engagement with the cover alignment features **204** and baseplate alignment features **210**.

(18) FIG. 2B shows a plan view of cover **202** when spacer **214** is installed onto cover **202**. The spacing contacts **216** each surround one of cover alignment features **204**. Frame **218** surrounds an exterior perimeter of the cover sealing surface **206** in the plan view of FIG. 2B, with an entirety of

the frame **218** being outside the cover sealing surface **206**, such that no part of frame **218** overlaps with the cover sealing surface **206**. Baseplate sealing surface **212** can be positioned to correspond to cover sealing surface **206** and thus have the same relationship to frame **218** as baseplate sealing surface **212**.

(19) FIG. 3 shows a spacer according to an embodiment. Spacer **300** includes spacing contacts **302** and frame **304**. Frame **304** includes spring arms **306**. Spacing contacts **302** are configured to contact alignment features of a cover and a baseplate of a reticle container, such as cover alignment features **204** and baseplate alignment features **210** shown in FIG. 2 and described above. The spacing contacts **302** can be located relative to one another such that spacing contacts **302** are spaced apart by approximately the same distances as the cover and baseplate alignment features. The frame **304** can join two or more of the spacing contacts **302**. The frame **304** can be configured to be outside cover and/or baseplate sealing surfaces such as sealing surfaces when spacer **300** is installed into a reticle pod and the reticle pod is viewed in plan view. The frame **304** can include spring arms **306**. Spring arms **306** can be features of frame **304** allowing deflection of the frame **304** such that the relative position of spacing contacts **302** can be varied. The spring arms can be segments that vary in thickness, include bends or other such features, and/or include resilient materials such that the frame can be deflected at those spring arms **306**. Any suitable number of spring arms **306** can be included in frame **304**. In an embodiment, a spring arm **306** is provided in frame **304** for each of the spacing contacts **302** included in spacer **300**. In an embodiment, the number of spacing contacts **302** included in spacer **300** is greater than the number of spring arms included in frame **304**.

(20) FIG. 4 shows a cover and a spacer according to an embodiment. Cover **400** includes cover flange **402**. Spacer **404** includes spacing contacts (not shown) joined by a frame **406**, and lifting arms **408**. The spacing contacts can be any of the spacing contacts described above. The frame **406** can join two or more of the spacing contacts. The frame **406** can be configured to be outside cover and/or baseplate sealing surfaces when spacer **404** is installed into a reticle pod including cover **400** and the reticle pod is viewed in plan view. Lifting arms **408** can be provided on spacer **402**, for example extending from the frame **406**. Lifting arms **408** are positioned such that lifting contacts **410** can contact the cover or the baseplate of a reticle pod when the spacer **400** is installed into the reticle pod. The lifting contacts **410** can contact cover **400**, for example at the cover flanges **402**. The lifting contacts **410** can be on flexible arms **412** included in lifting arms **408**. The flexible arms **412** can have a flexibility selected such that the lifting arms **408** can support the cover **400** apart from a baseplate when spacer **404** is installed into the inner pod of a reticle pod by itself. The flexible arms can be configured to deflect when the cover and baseplate are pressed together, for example when the cover and the baseplate are placed within an outer pod such as outer pod **102** described above and shown in FIG. 1. The flexible arms can result in there being a first gap between the sealing surfaces of the cover **400** and the baseplate when an inner pod having spacer **404** installed is contained within an outer pod of a reticle pod, and a second gap, larger than the first, when the inner pod having spacer **404** installed is removed from the outer pod but the cover **400** and baseplate remain together. The larger second gap can reduce the likelihood of contact between cover **400** and baseplate sealing surfaces during unpacking of the inner pod and removal of the spacer **404**, for example following shipment of a reticle pod in which spacer **404** is used to separate the cover **400** and the baseplate during shipping.

(21) FIG. 5 shows a reticle pod according to an embodiment. Reticle pod **500** includes outer pod **502** including pod dome **504** and pod door **506**. Reticle pod **500** further includes inner pod **508**, which includes cover **510** and baseplate **512**. Cover **510** includes cover sealing surface **514** and cover flanges **518**. Baseplate **512** includes baseplate sealing surface **516**. Spacer **520** can be installed into the outer pod **502** and engage with inner pod **508** so as to provide a gap between the cover sealing surface **514** and baseplate sealing surface **516**. The spacer **520** can include a first spacer frame **522**, a second spacer frame **524**, anchors **526**, and cover supports **528**. The anchors

526 can attach to support posts **530** included in pod door **506**.

(22) Outer pod **502** is an outer pod of reticle pod **500** configured to define an internal space capable of accommodating inner pod **508**. Outer pod **502** includes pod dome **504** and pod door **506**, which combine to define the internal space. Pod door **506** can be joined to pod dome **504** by any suitable connection, such as latching mechanisms or the like. The pod door **506** can include support posts **530**. Support posts **530** can be configured to contact the inner pod **508**, for example at baseplate **512**, so as to position the inner pod **508** within the internal space of outer pod **502**.

(23) Inner pod **508** is an inner pod of reticle pod **500** configured to accommodate a reticle. Inner pod **508** includes cover **510** and baseplate **512**. Cover **510** and baseplate **512** can include any suitable features of covers and baseplates included herein, such as, as a non-limiting example, the cover and baseplate alignment features **204**, **210** as discussed above and shown in FIGS. **2A** and **2B**. Cover **510** and baseplate **512** respectively include cover sealing surface **514** and baseplate sealing surface **516**. The cover sealing surface **514** and the baseplate sealing surface **516** are surfaces respectively provided on cover **510** and baseplate **512** so as to form a seal when the inner pod **508** is assembled in the absence of spacer **520**, for example, when inner pod **508** is accommodating a reticle. The sealing can be a seal based on contact of the cover and baseplate sealing surfaces **514**, **516**, close proximity allowing generation of a pressure differential between an inside and an exterior of the inner pod **508**, or any other suitable sealing of the inner pod **508**.

When spacer **520** is used in reticle pod **500**, the cover sealing surface **514** and baseplate sealing surface **516** can be spaced apart by a gap. The gap can be maintained by the presence of spacer **520** and its engagement with parts of reticle pod **500**, such as the pod dome **504**, pod door **506**, cover **510**, and/or baseplate **512**. The gap can have any suitable size to reduce or eliminate contact between the cover and baseplate sealing surfaces **514**, **516** while allowing the inner pod **508** to still be accommodated within the outer pod **502**. In an embodiment, the gap can have a size of between 20 micrometer (μm) and 1 millimeter (mm). Spacer **520** can be installed into reticle pod **500** to provide and maintain the gap, for example, during transportation and/or storage of the reticle pod **500**. The maintenance of the gap can reduce or eliminate rubbing friction, and in turn reduce wear and accompanying particle generation at the inner pod **508**. The maintenance of the gap can also reduce the likelihood or extent of damage resulting from contact between the cover **510** and baseplate **512** due to shocks occurring during transportation or storage of reticle pod **500**.

(24) In the embodiment shown in FIG. **5**, the cover **510** includes cover flanges **518**. The cover flanges **518** can project from opposing sides of the cover **510**. The cover flanges **518** can be contacted by one or more features of the spacer **520** such that the cover **510** can be retained in a particular position within outer pod **502**, for example supporting cover **510** in a particular position so as to provide the gap between cover sealing surface **514** and baseplate sealing surface **516**.

(25) Spacer **520** can be installed into the reticle pod **500** to maintain the gap between cover sealing surface **514** and baseplate sealing surface **516**, for example to prevent wear or damage during the transportation and/or storage of reticle pod **500**. Spacer **520** can engage with the cover **510** and baseplate **512** so as to retain the cover **510** and baseplate **512** such that the gap is maintained when reticle pod **500** is assembled. The spacer **520** can include one or more discrete structures. The spacer **520** or portions thereof can be configured to engage with pod dome **504** and/or pod door **506** so as to fix the position of spacer **520**.

(26) First spacer frame **522** is a frame configured to be outside of the cover sealing surface **514** and the baseplate sealing surface **516** when reticle pod **500** is assembled. First spacer frame **522** can include features configured to engage with is configured to position engagement features such that the engagement features can contact one of cover **510** or baseplate **512** so as to position the cover **510** or baseplate **512** such that the gap is maintained when spacer **520** is installed into reticle pod **500**. The engagement features can be, for example, flanges, tabs, spacing contacts such as spacing contacts **216** described above and shown in FIG. **2B**, or the like.

(27) Second spacer frame **524** is a frame configured to be outside of the cover sealing surface **514**

and the baseplate sealing surface **516** when reticle pod **500** is assembled. Second spacer frame **524** is configured to position engagement features such that the engagement features can contact one of cover **510** or baseplate **512** so as to position the cover **510** or baseplate **512** such that the gap is maintained when spacer **520** is installed into reticle pod **500**. In the embodiment shown in FIG. 5, the engagement features positioned by the second spacer frame **524** can be cover supports **528**. The cover supports **528** can be projections configured to contact cover flanges **530** to support cover **510** above baseplate **512** so as to maintain the gap between cover sealing surface **514** and baseplate sealing surface **516**. In other embodiments, the engagement features can be, for example, flanges, tabs, spacing contacts such as spacing contacts **216** described above and shown in FIG. 2B, or the like. In an embodiment, the second spacer frame **524** is part of a continuous structure with first spacer frame **522**. In an embodiment, second spacer frame **524** is part of a separate structure from the structure including first spacer frame **522**.

(28) Anchors **526** are used to connect spacer **520** or portions thereof to at least one of the pod dome **504** or pod door **506**. The anchors can be configured to engage with any suitable features of the pod dome **504** or pod door **506** such that the position of the spacer **520** or portions thereof such as the first spacer frame **522** and/or second spacer frame **524** can be maintained. In the embodiment shown in FIG. 5, the anchors **526** are configured to engage with support posts **530** provided on the pod door **506**. The anchors **526** can be configured to engage with features inherently included in the pod dome **504** or pod door **506** and/or features specifically included in the pod dome **504** or pod door **506** for engagement with anchors **526**. The features can include any mechanical engagement features, such as posts, flanges, grooves, ledges, shoulders, and the like. In an embodiment, a first one or more anchors **526** are configured to position the first spacer frame **522**, and a second one or more anchors **526** are configured to position the second spacer frame **524**.

(29) Aspects:

(30) It is understood that any of aspects 1-7 can be combined with any of aspects 8-15 or aspects 16-20. It is understood that any of aspects 8-15 can be combined with any of aspects 16-20.

(31) Aspect 1. A spacer for a reticle pod, comprising: a plurality of spacing contacts, each of the spacing contacts configured to contact a first reticle pod segment at a first reticle pod feature and to contact a second reticle pod segment at a second reticle pod feature, wherein the first reticle pod feature is outside of a first sealing surface of the first reticle pod segment, and the second reticle pod feature is outside of a second sealing surface of the second reticle pod segment; and a frame joining at least two of the plurality of spacing contacts, the frame configured to be outside the first sealing surface and the second sealing surface in plan view of the reticle pod, wherein the spacing contacts are configured to contact the first reticle pod segment and the second reticle pod segment such that the first reticle pod segment and the second reticle pod segment are separated by a gap.

(32) Aspect 2. The spacer according to aspect 1, wherein the plurality of spacing contacts includes at least three of the spacing contacts.

(33) Aspect 3. The spacer according to any of aspects 1-2, wherein the frame joins all of the spacing contacts of the plurality of spacing contacts.

(34) Aspect 4. The spacer according to any of aspects 1-3, wherein the frame comprises at least one spring arm configured to allow deflection of the frame such that a distance between the plurality of spacing contacts is varied.

(35) Aspect 5. The spacer according to any of aspects 1-4, wherein each of the spacing contacts includes an inner surface defining an opening configured to receive the first reticle pod feature, and an outer surface configured to be received within the second reticle pod feature.

(36) Aspect 6. The spacer according to any of aspects 1-5, wherein the gap has a width in a range from 20 μm to 1 mm.

(37) Aspect 7. The spacer according to any of aspects 1-6, wherein the spacer includes at least one of polycarbonate and a polyolefin.

(38) Aspect 8. A reticle pod, comprising: an inner pod including a cover and a baseplate, the cover

including a cover sealing surface and a plurality of cover features outside of the cover sealing surface, the baseplate including a baseplate sealing surface and a plurality of baseplate features outside of the baseplate sealing surface; and a spacer, the spacer including: a plurality of spacing contacts, each of the spacing contacts configured to contact one of the plurality of cover features and to contact one of the plurality of baseplate features; and a frame joining at least two of the plurality of spacing contacts, the frame configured to be outside the cover sealing surface and the baseplate sealing surface in plan view of the reticle pod, wherein the spacing contacts are configured to contact the cover and the baseplate such that the cover and the baseplate are separated by a gap.

(39) Aspect 9. The reticle pod according to aspect 8 wherein the plurality of spacing contacts includes at least three of the spacing contacts.

(40) Aspect 10. The reticle pod according to any of aspects 8-9, wherein the frame joins all of the spacing contacts of the plurality of spacing contacts.

(41) Aspect 11. The reticle pod according to any of aspects 8-10, wherein the frame comprises at least one spring arm configured to allow deflection of the frame.

(42) Aspect 12. The reticle pod according to any of aspects 8-11, wherein the cover feature is a pin and the baseplate feature is a channel configured to receive the pin, the channel including a lead-in.

(43) Aspect 13. The reticle pod according to aspect 12, wherein each of the spacing contacts includes an inner surface defining an opening configured to receive the pin, and an outer surface configured to be received within the lead-in.

(44) Aspect 14. The reticle pod according to any of aspects 8-13, wherein the gap has a width in a range from 20 μm to 1 mm.

(45) Aspect 15. The reticle pod according to any of aspects 8-14, further comprising an outer pod including a pod dome and a pod door, the outer pod configured to accommodate the inner pod.

(46) Aspect 16. A method of preparing a reticle pod for shipment, comprising: assembling an inner pod including a cover, a baseplate, and a spacer provided between the cover and the baseplate, the spacer including a plurality of spacing contacts contacting the cover at cover features outside of a cover sealing surface and the spacer contacting the baseplate at baseplate features outside of a baseplate sealing surface and a frame joining at least two of the plurality of spacing contacts; wherein the cover and the baseplate are separated by a gap.

(47) Aspect 17. The method according to aspect 16, wherein the frame includes a deflection arm, and assembling the inner pod includes deflecting the frame at the deflection arm to align at least one of the plurality of spacing contacts with one of the cover features or one of the baseplate features.

(48) Aspect 18. The method according to any of aspects 16-17, wherein the gap has a width in a range from 20 μm to 1 mm.

(49) Aspect 19. The method according to any of aspects 16-18, wherein the cover features are each a pin, the baseplate features are each a channel including a lead-in, and assembling the inner pod includes passing each of the pins through one of the spacing contacts, and inserting each of the spacing contacts into one of the lead-ins.

(50) Aspect 20. The method according to any of aspects 16-19, further comprising: placing the inner pod into an outer pod, the outer pod including a pod dome and a pod door; joining the pod dome to the pod door to enclose the inner pod; and shipping the reticle pod from a packing location to a destination location while the pod dome is joined to the pod door.

(51) The examples disclosed in this application are to be considered in all respects as illustrative and not limitative. The scope of the invention is indicated by the appended claims rather than by the foregoing description; and all changes which come within the meaning and range of equivalency of the claims are intended to be embraced therein.

Claims

1. A spacer for a reticle pod, comprising: a plurality of spacing contacts, each of the spacing contacts configured to contact a first reticle pod segment at a first reticle pod feature and to contact a second reticle pod segment at a second reticle pod feature, wherein the first reticle pod feature is outside of a first sealing surface of the first reticle pod segment, and the second reticle pod feature is outside of a second sealing surface of the second reticle pod segment; and a frame joining at least two of the plurality of spacing contacts, wherein the spacing contacts are configured to contact the first reticle pod segment and the second reticle pod segment such that the first reticle pod segment and the second reticle pod segment are separated by a gap.
2. The spacer of claim 1, the frame is configured to be outside the first sealing surface and the second sealing surface.
3. The spacer of claim 1, wherein the frame joins all of the spacing contacts of the plurality of spacing contacts.
4. The spacer of claim 1, wherein the frame comprises at least one spring arm configured to allow deflection of the frame such that a distance between the plurality of spacing contacts is varied.
5. The spacer of claim 1, wherein each of the spacing contacts includes an inner surface defining an opening configured to receive the first reticle pod feature, and an outer surface configured to be received within the second reticle pod feature.
6. The spacer of claim 1, further comprising lifting arms extending from the frame wherein the lifting arms include flexible arms with lifting contacts.
7. A reticle pod, comprising: an inner pod including a cover and a baseplate, the cover including a cover sealing surface and a plurality of cover features outside of the cover sealing surface, the baseplate including a baseplate sealing surface and a plurality of baseplate features outside of the baseplate sealing surface; and a spacer, the spacer including: a plurality of spacing contacts, each of the spacing contacts configured to contact one of the plurality of cover features and to contact one of the plurality of baseplate features; and a frame joining at least two of the plurality of spacing contacts, wherein the spacing contacts are configured to contact the cover and the baseplate such that the cover and the baseplate are separated by a gap.
8. The reticle pod of claim 7 wherein the frame is configured to be outside the cover sealing surface and the baseplate sealing surface.
9. The reticle pod of claim 7, wherein the frame comprises at least one spring arm configured to allow deflection of the frame.
10. The reticle pod of claim 7, wherein the cover feature is a pin and the baseplate alignment is a channel configured to receive the pin, the channel including a lead-in.
11. The reticle pod of claim 10, wherein each of the spacing contacts includes an inner surface defining an opening configured to receive the pin, and an outer surface configured to be received within the lead-in.
12. The reticle pod of claim 7, wherein the spacer includes lifting arms extending from the frame.
13. The reticle pod of claim 12, wherein the lifting arms include flexible arms with lifting contacts positioned such that the lifting contacts can contact the cover.
14. The reticle pod of claim 7, further comprising an outer pod including a pod dome and a pod door, the outer pod configured to accommodate the inner pod.
15. A method of preparing a reticle pod for shipment, comprising: assembling an inner pod including a cover, a baseplate, and a spacer provided between the cover and the baseplate, the spacer including a plurality of spacing contacts contacting the cover at cover features outside a cover sealing surface and the spacer contacting the baseplate at baseplate features outside a baseplate sealing surface and a frame joining at least two of the plurality of spacing contacts; wherein the cover and the baseplate are separated by a gap.

16. The method of claim 15, wherein the frame includes a deflection arm, and assembling the inner pod includes deflecting the frame at the deflection arm to align at least one of the plurality of spacing contacts with one of the cover features or one of the baseplate features.
17. The method of claim 15, wherein the spacer includes lifting arms extending from the frame.
18. The method of claim 17, wherein the lifting arms include flexible arms with lifting contacts positioned such that the lifting contacts can contact the cover.
19. The method of claim 16, wherein the cover features are each a pin, the baseplate features are each a channel including a lead-in, and assembling the inner pod includes passing each of the pins through one of the spacing contacts, and inserting each of the spacing contacts into one of the lead-ins.
20. The method of claim 16, further comprising: placing the inner pod into an outer pod, the outer pod including a pod dome and a pod door; joining the pod dome to the pod door to enclose the inner pod; and shipping the reticle pod from a packing location to a destination location while the pod dome is joined to the pod door.
-